Alt Leaf Gate vs Leaf+Pest Gate — Comparison

Same TinyEngine student and training recipe · different decision: binary leaf/not_leaf versus 3-class leaf/others/pest

This report compares the two field-hardened gates that currently sit in *Ikmal/Gate Model*: the **Alt Leaf Gate** (*aclis_leaf_gate_96x_alt_full_int8.tflite*) and the **Leaf+Pest Gate** (*aclis_pest_gate_96x_full_int8.tflite*). They share a MCU-sized depthwise-separable backbone and the same distillation + MixUp + EMA + camera-aug + QAT recipe. They do **not** solve the same classification problem, so overall accuracy must be read next to the class set, the cascade role, and the camera-stress split. All figures are taken from the two model folders (including the per-model *results.pdf* files generated from those same sources).

Sources used

Alt: Leaf Gate Model/Alt Leaf Gate Model/ALT_TRAINING_REPORT.md, alt_training_record.pdf, results.pdf, aclis_leaf_gate_alt_distill_qat.ipynb. Pest: Leaf+Pest Gate Model/training_record.pdf, aclis_pest_gate_distill_qat.ipynb, leaf_pest_others_dataset/ counts, results.pdf. Architecture notes: pest notebook states it is adapted from the alt recipe with the same student topology and a 3-way head.

1. What each model is for

The alt leaf gate is a binary filter in front of the PlantVillage disease CNN: continue if the frame is a leaf, otherwise reject. The pest gate keeps that leaf path and adds a third exit for PlantVillage pest images, so non-leaf clutter (others) and pests are no longer one bucket. Flash cost is essentially identical (~51 KB), so the choice is about **capability versus leaf-path reliability**, not MCU budget.

	Alt Leaf Gate	Leaf+Pest Gate
Role in cascade	capture → leaf / not_leaf → if leaf: disease CNN	capture → leaf / others / pest → leaf: disease CNN; pest: pest path; others: reject
Classes	leaf, not_leaf	leaf, others, pest
Student	TinyLeafGate, 28,854 params	TinyPestGate, 28,883 params (same blocks, 3-way head)
Teacher (train only)	MobileNetV3-Small	MobileNetV3-Small
Recipe	KD + MixUp + EMA + field augs + QAT	Same recipe, adapted from the alt notebook
INT8 artifact	51.1 KB	51.0 KB
Disease + gate flash	774.1 / 1024 KB (FITS)	774.0 / 1024 KB (FITS)
Input	96×96 RGB, full-frame Resize(96)	96×96 RGB, full-frame Resize(96)
Notebook target	≥ 92% (PASS)	≥ 90% (PASS)

Table 1. Shared MCU footprint; different decision and class set.

2. Datasets are related, not identical

Pest-gate *leaf* and *others* are the same Kaggle leaf / non-leaf images used by the alt gate. Pest-gate then **adds** PlantVillage pest images as a third class. That makes the pest-gate test set larger (4,319 vs 3,673) and the decision harder: damaged leaf tissue can look like pest, which shows up later as leaf→pest swaps.

Split	Alt leaf	Alt not_leaf	Alt total	Pest leaf	Pest others	Pest pest	Pest total
train	9,065	8,066	17,131	9,065	8,066	3,009	20,140
val	1,942	1,728	3,670	1,942	1,728	644	4,314
test	1,943	1,730	3,673	1,943	1,730	646	4,319

Table 2. Test leaf and others counts match across models; pest is extra.

3. Overall accuracy

Headline INT8 accuracy is **not directly comparable**: alt is a 2-way problem on a saturated Kaggle split; pest is a 3-way problem that includes the harder pest class. Both beat their own notebook targets. The useful comparison is (a) how each behaves on the shared leaf images, and (b) how much each drops under the same style of camera-stress transform.

Alt clean INT8	Pest clean INT8	Alt stress INT8	Pest stress INT8
99.07%	**95.58%**	**98.86%**	**91.06%**
2-class · n = 3,673	3-class · n = 4,319	drop 0.21 pts	drop 4.52 pts

Metric	Alt Leaf Gate	Leaf+Pest Gate	Read as
Clean INT8 overall	99.07%	95.58%	Different tasks; both PASS their target
Clean FP32 overall	~99.1–99.5%	95.58%	INT8 matches float in both records
|INT8 – FP32|	~0.03 pts	0.00 pts	Quantisation is not the difference
INT8 ↔ PT agreement	~99.9%	99.84%	Exported graph tracks PyTorch
Camera-stress INT8	98.86%	91.06%	Alt stays near ceiling; pest drops below 92%
Best EMA val	99.10%	95.58%	Student distillation checkpoint
Teacher best val	~99.92%	99.21%	Train-only MobileNetV3-Small

Table 3. Overall numbers from each folder's INT8 test. Camera-stress is the same idea (blur + JPEG + underexposure) in both notebooks.

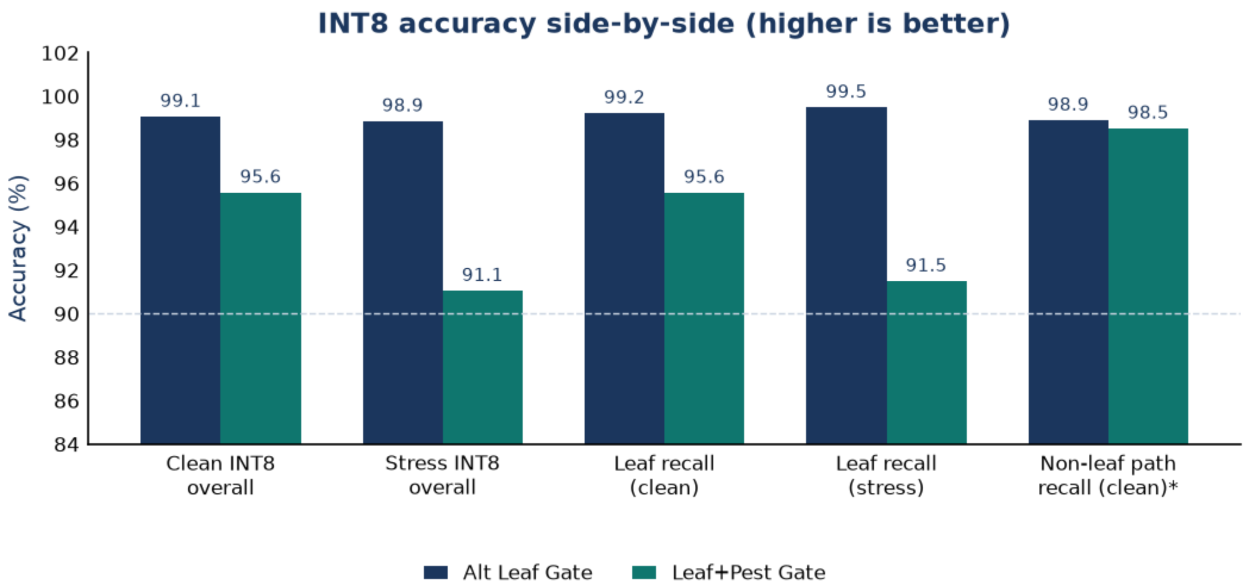

*Figure 1. *Non-leaf path for pest gate is others+pest collapsed to "not the disease CNN" (2,341 / 2,376 = 98.53%). Alt non-leaf path is not_leaf recall.*

4. Per-class accuracy

Per-class accuracy is recall. The only class both models share is **leaf**. On clean INT8 the alt gate keeps 99.23% of true leaves on the disease path; the pest gate keeps 95.57%. That 3.66-point gap is almost entirely the 75 true leaves predicted as pest. Pest itself is the pest gate's strongest class (96.13% clean, 95.4% under stress).

Class	Alt clean	Alt stress	Pest clean	Pest stress	What an error does
leaf	99.23% (1,928/1,943)	99.49% (1,933/1,943)	95.57% (1,857/1,943)	91.5%	Missed leaf never reaches the disease CNN
not_leaf / others	98.90% (1,711/1,730)	98.15% (1,698/1,730)	95.38% (1,650/1,730)	89.0%	Clutter sent to leaf or pest path
pest	— (no class)	—	96.13% (621/646)	95.4%	Pest sent to disease CNN or rejected
Overall	99.07%	98.86%	95.58%	91.06%	All test images

Table 4. Recall. Alt stress leaf actually rises slightly (99.23 → 99.49%); pest-gate leaf falls 4.1 points under the same style of degradation.

5. Confusion matrices (clean INT8)

Rows are true labels; columns are predictions. Diagonal cells are correct. Alt's 34 mistakes are a handful of leaf↔not_leaf swaps. Pest's 191 mistakes are dominated by **leaf → pest (75)** and **others → pest (52)** — the new third class absorbs most of the extra error.

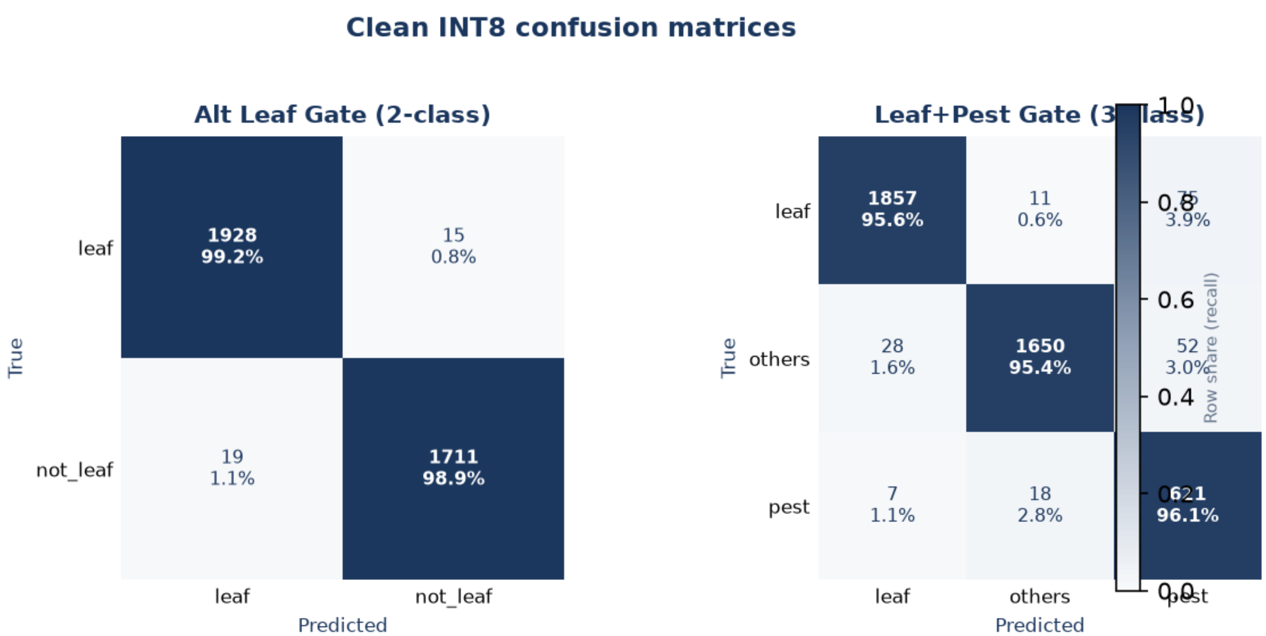

Figure 2. Clean INT8 matrices from alt FN/FP counts and from pest training_record.pdf §8.

True → pred	Alt count	Pest count	Cascade meaning
leaf → leaf	1,928	1,857	Disease CNN runs (desired for a real leaf)
leaf → not_leaf / others	15	11	Frame rejected; disease CNN skipped
leaf → pest	n/a	75	New alt-vs-pest failure: leaf diverted to pest path
not_leaf/others → leaf	19	28	Junk / clutter enters disease CNN
pest → leaf	n/a	7	Pest frame enters disease CNN
pest → pest	n/a	621	Pest path taken correctly

Table 5. Shared error types, plus the pest-only leaf→pest diversion.

6. Camera-stress robustness

Both notebooks evaluate a deterministic camera-stress proxy (full-frame resize + blur + underexposure + JPEG) meant to stand in for OV2640 / Arducam frames. Alt was trained for that gap and barely moves (99.07% → 98.86%). Pest uses the same recipe but the 3-way boundary is less stable: 95.58% → 91.06%. Others is the weak stress class (89.0%); pest class itself stays high (95.4%).

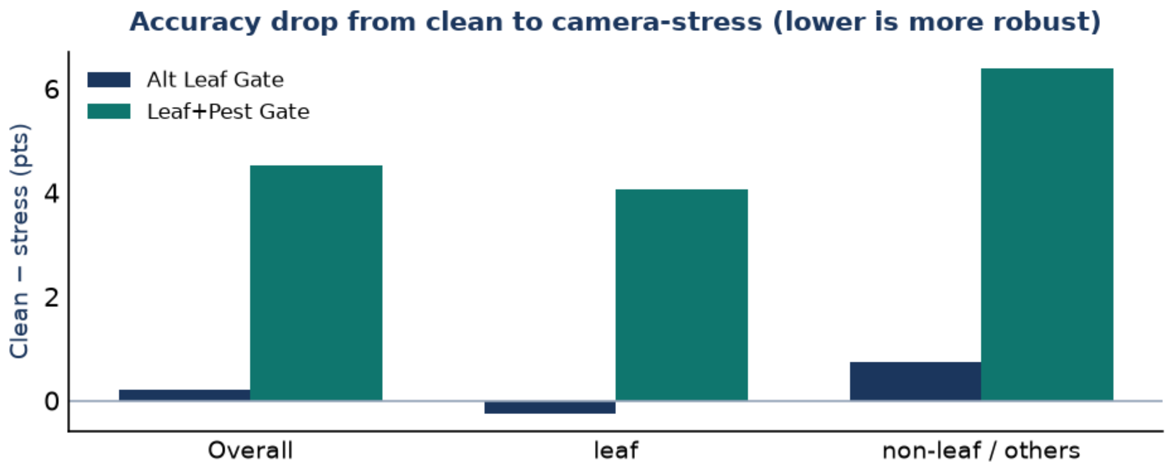

Figure 3. Positive bars = accuracy lost under stress. Alt leaf recall improves slightly (negative bar).

7. If the only question is "does a leaf reach the disease CNN?"

Collapsing pest-gate others+pest into a single "not the disease path" makes the two models comparable as leaf filters. On the same 1,943 test leaves, alt sends 1,928 to the disease CNN and pest sends 1,857. False leaves into the disease CNN stay small on both (19 vs 35, the extra 16 coming from others and pest). So: pest gate buys a working pest exit (621 / 646) at the cost of 75 true leaves that no longer reach the disease model.

Leaf-filter view (clean INT8)	Alt Leaf Gate	Leaf+Pest Gate (collapsed)
True leaf → disease CNN	1,928 / 1,943 (99.23%)	1,857 / 1,943 (95.57%)
True leaf missed	15	86 (11 others + 75 pest)
Non-leaf → disease CNN (FP)	19 / 1,730 (1.10%)	35 / 2,376 (1.47%)
Pest recognised as pest	not modelled	621 / 646 (96.13%)

Table 6. Same 1,943 Kaggle leaves in both tests; pest-gate non-leaf denominator is others+pest.

8. Conclusion

Use the Alt Leaf Gate when the cascade only needs a reliable leaf / not-leaf decision in front of the disease CNN. It is near ceiling on clean Kaggle data and stays there under the camera-stress proxy (98.86% overall, 99.5% leaf recall), at 51.1 KB.

Use the Leaf+Pest Gate when the product must branch pest frames away from both the disease CNN and a generic reject. It does that well (pest recall 96.1% clean / 95.4% stress) at the same flash cost, but overall accuracy is lower because the task is 3-way, leaf recall falls to 95.6% clean and 91.5% under stress, and 75 true leaves are labelled pest.

They are not competing on the same scoreboard. Alt is the stronger *leaf filter*; pest is the only model that can *name a pest*. A field A/B on STM32 + OV2640 should score three things: (1) true leaves still entering the disease CNN, (2) pest frames taking the pest path, (3) clutter not wasting a disease-CNN call.